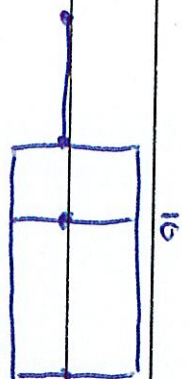


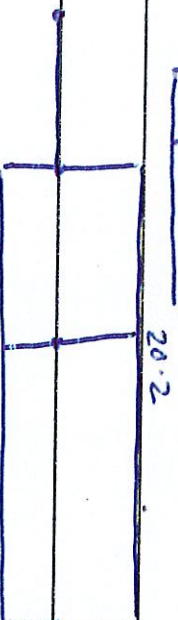
Core Data Practice SAC 4.

Part 1

(A)



WILD ANIMALS.



MAMMALS.



LIFE SPAN.

- consider labelled an accurate axis
- label each box plot
- instructions for CAS - how to create 5 number summary + box plot.

* other types of graphical displays include:

~~box~~ dot plot, histogram, stem + leaf.

Discussing the data

- consider the shape, outliers, range
- eg. The ^{parallel} box plot above shows that the life span for wild animals is positively skewed with no outliers, it has a range of $(30 - 3 = 27)$ 27 and a median of 10.

The life span of mammals is also positively skewed with a larger range of $(46 - 7.6 = 38.4)$ 38.4 with no outliers and a median of 20.

(B) Does an association exist?

- An association exists if there is a difference in medians (can also consider IQR & range, but only need to discuss 1.)
- clearly state the variables

The median life span of mammals is 20.2 yrs whilst the median life span of wild animals was 10 yrs. This confirms an association between life span and the type of animal.

c. Do you have instructions for calculating mean + sd in your exam book?

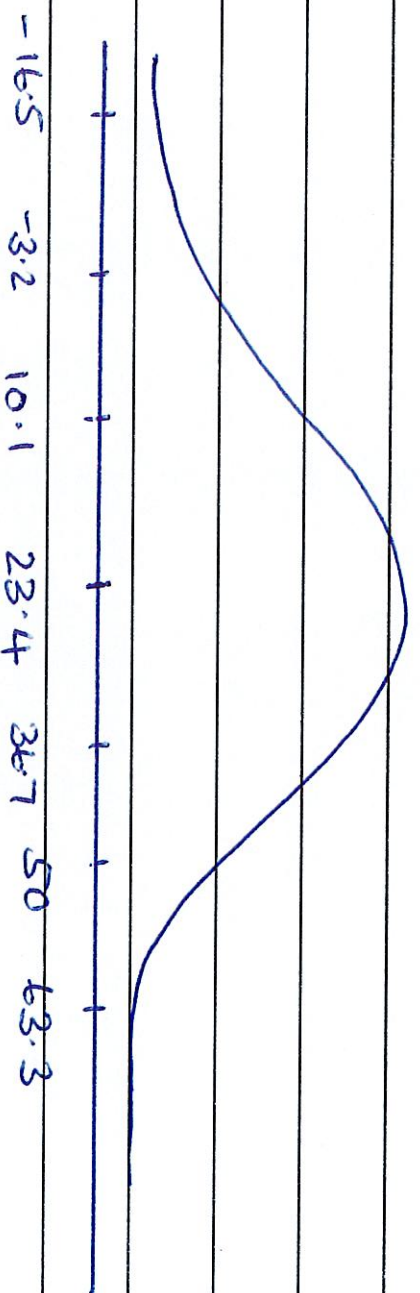
• A sentence to 'interpret the 2-score'.

ONE OF

MAMMALS.

mean = 23.4

sd = 13.3



LIFE SPAN MAMMALS.

$z = 2.1$

$z \text{ score} = \frac{\text{actual} - \text{mean}}{\text{sd}} =$

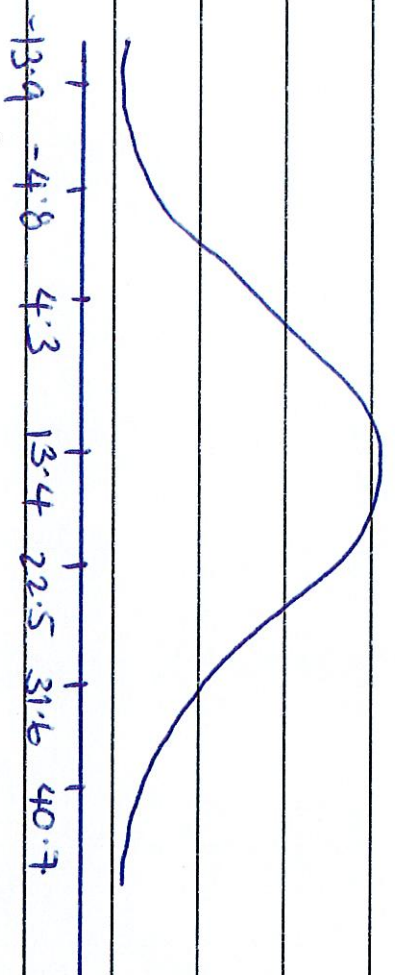
$$\text{solve } \left(\frac{2.1 = \text{actual} - \frac{23.4}{13.3}}{13.3} \right) = 51.33$$

A z-score of 2.1 suggests this animal has a life span of 51.33 yrs and is in the top 2.5% of life spans of mammals.

WILD ANIMALS.

mean = 13.4

sd = 9.1



The z score of 2.1 suggests the animal has a life span of 32.51 yrs and is in the top 2.5% of life spans of wild animals.

Regression Analysis

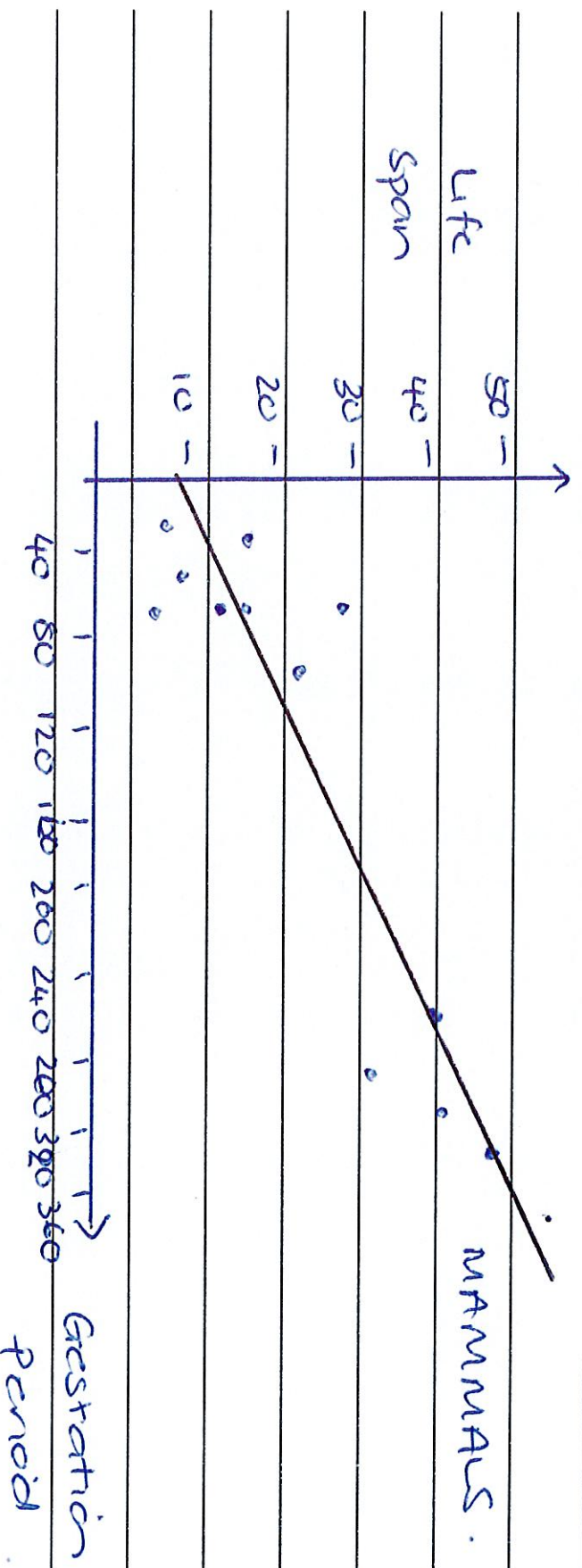
Part 2 : Life Span Mammal. is predicted from the gestation period

• do you have instructions in your exam book on what/how to complete Regression Analysis -including instructions for CAS.

- do you have the sentence structure for Slope, Intercept, coefficient of determination.
- must use the variables you are provided, NOT $x + y$.

A. Scatterplot - use a ruler, graph paper.
- label axis + heading.

- include LSR



Correlation coefficient, r

$r = 0.895$ The correlation coefficient indicates a strong, positive linear graph.

Least sq. Regression Line

Life span = $9.68 + 0.1 \times$ Gestation Period.

Coefficient of determination, r^2

$$r^2 = 0.8 = 80\%$$

The C.O.D. tells us that 80% of the variation in the life span can be explained by the variation in the gestation period.

$$\text{slope} = b$$

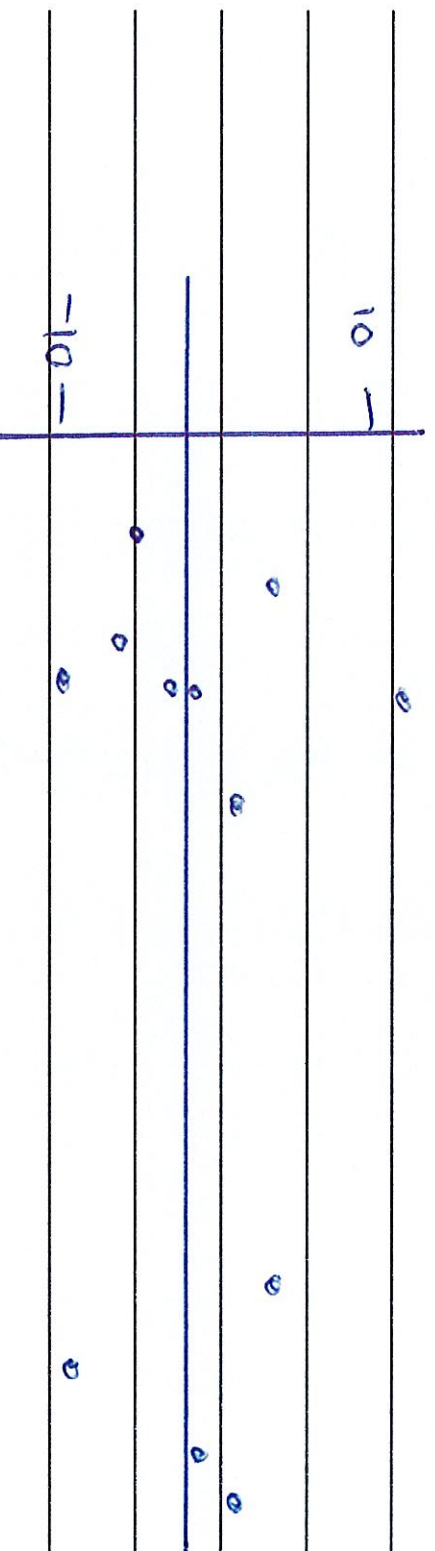
The slope indicates that the life span increases by 0.1 yr for every 1 day of gestation period.

$$\text{Intercept} = a$$

When the gestation period is ~~9.68~~ days the life span will 9.68 yrs

(NOTE) This is an example of when the interpretation of the intercept is not valid

Residual plot



Residual plot shows no clear pattern suggesting a linear relationship between life span and gestation period of mammals

Prediction if gestation period = 52 days.

$$\text{Life span} = 9.68 + 0.1 \times 52 = 14.88 \text{ yrs.}$$

$$\text{Actual (from table of data)} = 9.8 \text{ yrs}$$

$$\text{Residual} = \text{actual} - \text{predicted}$$

$$= 9.8 - 14.88$$

$$= -5.08$$

Data is reliable since it is within the data range that was analysed (interpolation).

Prediction: Gestation = 21 days.

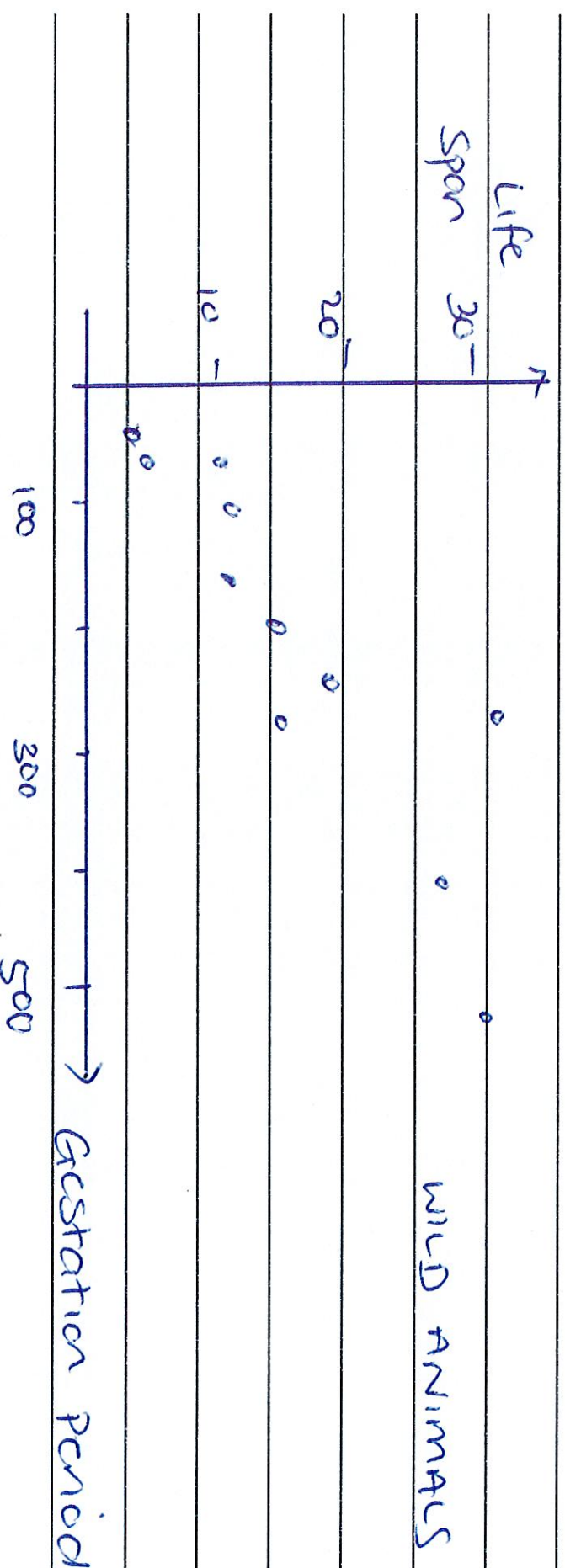
$$\text{Life span} = 9.68 + 0.1 \times 21 = 11.78 \text{ yrs.}$$

when gestation period is 21 days the predicted life span is 11.78 yrs, this data is not reliable as it is outside the analysed data range. (extrapolation).

Part 3: Transformation.

- Do you have the types of transformations graph in your exam book?
- Do you have clear instructions on how to transform data in CAS?
- Must use the variables provided - DO NOT use $x+y$.
- Don't forget to include the transformation in the eqn $\rightarrow$ the CAS will not remind you.

A. Possible Transformations:



Possible transformations are:

$(\text{Life span})^2$, $\log(\text{gestation period})$, $\frac{1}{\text{gestation}}$

	LSE.	r	r ²	Residue Plot.
$\log(gp)$	$ls = -22.47 + 17.74 \log(gp)$	0.85	0.73	No clear pattern.
$(ls)^2$	$(ls)^2 = -26.79 + 1.76 \times gp$	0.81	0.66	No clear pattern.

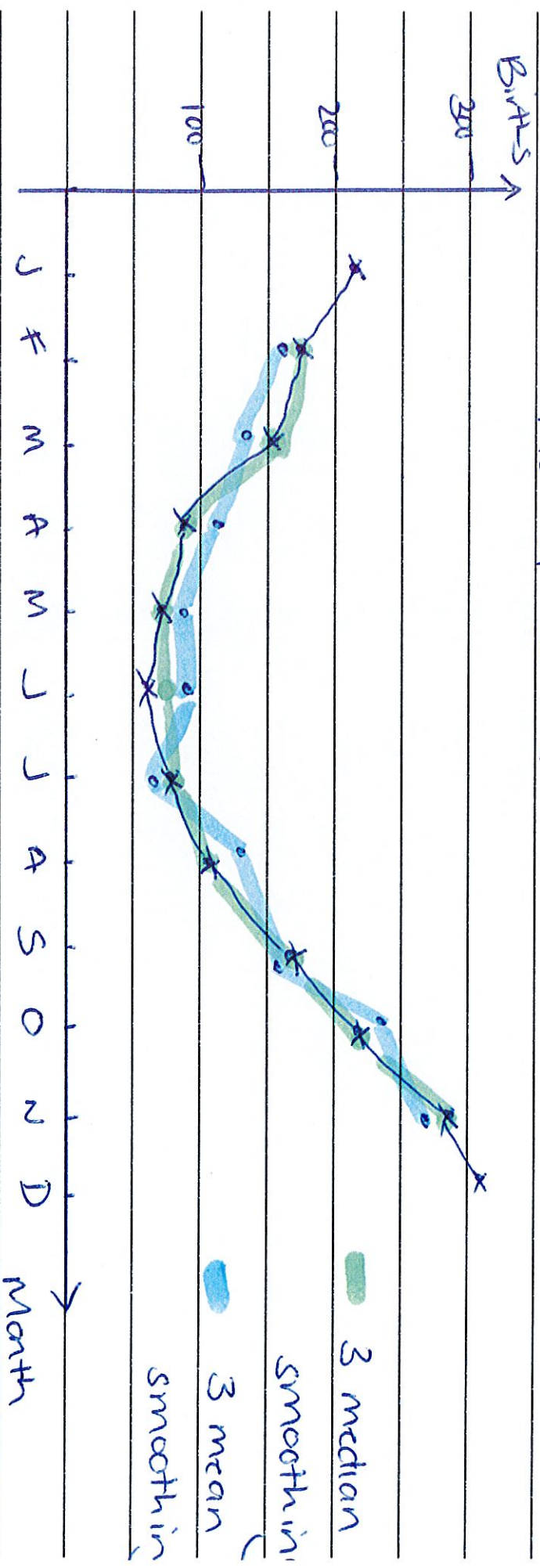
Best transformation has the largest C.O.D, r^2 , which in this case is $\log(\text{gestation period})$.

When $g.p = 75$ days

$$\begin{aligned} \text{life span} &= -22.47 + 17.74 \times \log(75) \\ &= 10.79 \text{ yrs.} \end{aligned}$$

Part 4: Time Series.

No. of Births.



The graph has a downward trend from Jan - June and then an increasing trend from July - Dec. There are ~~no patterns~~ ^{at} irregular fluctuations.

		3 mean smooth.	4 mean with cent
J	210		-
F	187	$\frac{210 + 187 + 156}{3}$	184
M	156	$\frac{187 + 156 + 93}{3}$	145
A	93	$\frac{156 + 93 + 85}{3}$	111
M	85	$\frac{93 + 85 + 62}{3}$	80.
J	62		$\frac{85 + 62 + 89 + 101}{4}$
J	89		79.
A	101		84
S	174		121
O	210		162
N	294		226
D	310.		$\frac{271 + 268}{2}$

Since there are not outliers, either mean smoothing or median smoothing is appropriate data smoothing.

4 mean smoothing with centering for July:

$$M, J, J, A = \frac{85 + 62 + 89 + 101}{4} \quad \swarrow$$

$$J, J, A, S = \frac{62 + 89 + 101 + 174}{4} \quad \searrow$$

$$\frac{84.25 + 106.5}{2} = 95.38$$

New value for July would be 95.